

EE115 Analog Circuits

OP AMP Basics

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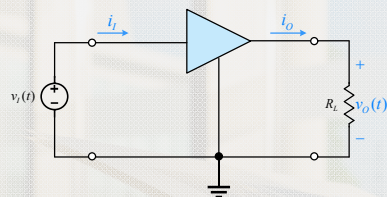
Outline

- OP AMP Basics
 - Amplifiers
 - Circuit Models for Amplifiers
 - Frequency Response of Amplifiers
- Reading: Sedra/Smith Book Pages 16-45



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Amplifier Gain



- Voltage Gain:

$$A_v = \frac{v_o}{v_i}$$

- Current Gain:

$$A_i = \frac{i_o}{i_i}$$

- Power Gain:

$$A_p = \frac{p_L}{p_i} = \frac{v_o i_o}{v_i i_i} = A_v A_i$$

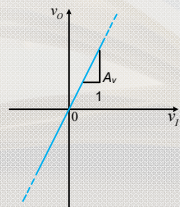
- Note: A_v and A_i can be **positive**, **negative**, or even **complex** numbers. Negative gain means the output is 180° out of phase with input. However, A_p should always be **positive**.

- Gain is usually expressed in **Decibel (dB)**

$$A_p(\text{dB}) = 10 \log |A_p|$$

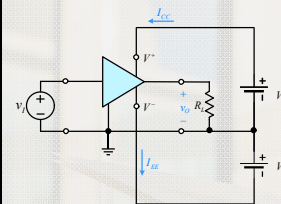
$$A_v(\text{dB}) = 10 \log |A_v|^2 = 20 \log |A_v|$$

$$A_i(\text{dB}) = 10 \log |A_i|^2 = 20 \log |A_i|$$



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Amplifier Power Supply and Dissipation



- Circuit needs dc **power supplies** to function
- Typical power supplies are denoted as V_{CC} (or V_{DD}) and $-V_{EE}$ (or V_{SS})

- Total **dc power dissipation**

$$P_{dc} = V_{CC} I_{CC} + V_{EE} I_{EE}$$

- Power balance

$$P_{dc} + P_i = P_L + P_{diss}$$

- P_i : power drawn from signal source
- P_L : power delivered to the load
- P_{diss} : power dissipated in the amplifier circuit

- Power efficiency

$$\eta = \frac{P_L}{P_{dc}}$$

- Important for power amplifiers: stereo output, transmitters

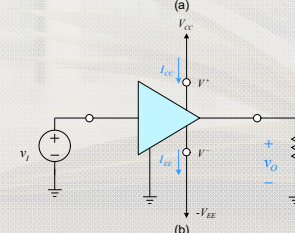


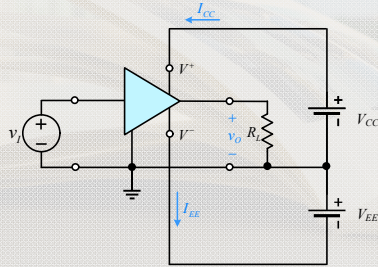
Fig. An amplifier with dual dc supplies.

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Amplifier Power Supply and Dissipation



- A microphone outputs a 0.4 V peak sinusoidal signal and delivers 100 μ A peak to an amplifier powered by ± 1 V supplies. The amplifier drives a 32 Ω speaker with a 0.8 V peak output and draws 30 mA from each supply.
- Find : Voltage gain; current gain; power gain; DC power supplied; power dissipation; and efficiency.



Solution:

$$A_v = \frac{0.8\text{V}}{0.4\text{V}} = 2\text{V/V, or } A_v = 20\log 2 = 6\text{ dB}$$

$$i_o = \frac{0.8\text{V}}{32\Omega} = 25\text{ mA}$$

$$A_i = \frac{i_o}{i_i} = \frac{25\text{ mA}}{0.1\text{ mA}} = 250\text{ A/A, or } A_i = 20\log 250 = 47.9\text{ dB}$$

$$P_L = V_{\text{rms}} I_{\text{rms}} = \frac{0.8\text{ V}}{\sqrt{2}} \frac{25\text{ mA}}{\sqrt{2}} = 10\text{ mW}$$

$$P_I = V_{\text{irms}} I_{\text{irms}} = \frac{0.4\text{ V}}{\sqrt{2}} \frac{0.1\text{ mA}}{\sqrt{2}} = 20\text{ }\mu\text{W}$$

$$A_p = \frac{P_L}{P_I} = \frac{10\text{ mW}}{20\text{ }\mu\text{W}} = 500\text{ W/W, or } A_p = 10\log 500 = 29.9\text{ dB}$$

$$P_{dc} = 1\text{V} \times 30\text{ mA} + 1\text{V} \times 30\text{ mA} = 60\text{ mW}$$

$$P_{\text{dissipated}} = P_{dc} + P_I - P_L = 60\text{ mW} + 0.02\text{ mW} - 10\text{ mW} \approx 50\text{ mW}$$

$$\eta = \frac{P_L}{P_{dc}} \times 100 = 16.7\%$$

Linear Range vs. Saturation Range

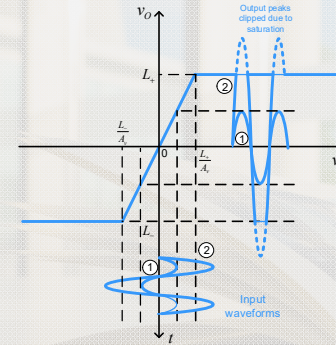


Fig. Typical amplifier transfer characteristics.

- **Within** linear range, the output voltage (or current) is **proportional** to the input voltage (or current)
- **Beyond** linear range, the output voltage (or current) waveforms **saturates**, resulting in distortions
 - Lose fidelity in stereo system

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Symbol Notation



$$i_c(t) = I_C + i_c(t)$$

where

- $i_c(t)$: Lowercase-uppercase $\rightarrow$ **total** current
- I_C : Uppercase-uppercase $\rightarrow$ **dc** current
- $i_c(t)$: Lowercase-lowercase $\rightarrow$ **small signal ac** current

$$i_c = I_C \sin(\omega t)$$

where,

- I_C : Uppercase-lowercase $\rightarrow$ **amplitude** of ac current (sinwave)

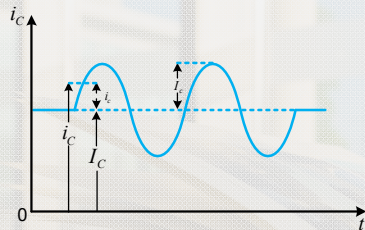


Fig. Symbol notation convention

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Circuit Model of Voltage Amplifiers



(Two-Port Model)

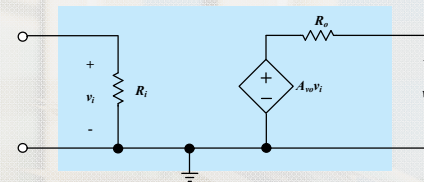


Fig. Circuit model for the voltage amplifier.

- A_{vo} : open-circuit voltage gain
- R_i : input resistance
- R_o : output resistance
- R_s : source resistance
- R_L : load resistance

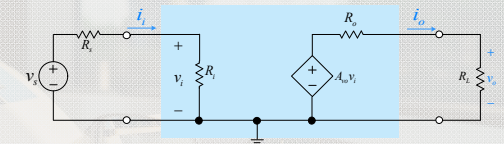


Fig. Voltage amplifier with input signal source and load.

For ideal op-amp, why we prefer large R_i and Small R_o ?

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Cascaded Amplifier

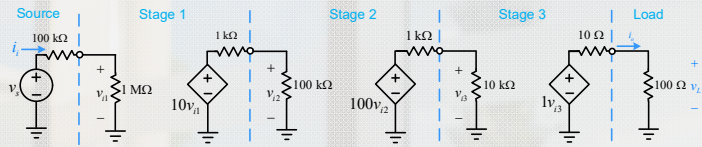


Fig. Three-stage amplifier

- For most practical applications, multiple stages of amplifiers are cascaded to
 - Provide sufficient **gain**
 - Provide adequate input and output **resistances/impedances**
- For example, in a voltage amplifier
 - The **input stage** is designed to have **high input impedance**
 - The **output stage** is designed to have **low output impedance**
 - **Middle stage** provides the necessary **gain**

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Amplifier Types



Table The Four Amplifier Types			
Type	Circuit Model	Gain Parameter	Ideal Characteristics
Voltage Amplifier		Open-Circuit Voltage Gain $A_{vo} = \frac{v_o}{v_i} _{R_L=\infty}$ (V/V)	$R_i = \infty$ $R_o = 0$
Current Amplifier		Short-Circuit Current Gain $A_{is} = \frac{i_o}{i_i} _{R_L=0}$ (A/A)	$R_i = 0$ $R_o = \infty$
Transconductance Amplifier		Short-Circuit Transconductance $G_{ms} = \frac{i_o}{v_i} _{R_L=0}$ (A/V)	$R_i = \infty$ $R_o = \infty$
Transresistance Amplifier		Open-Circuit Transresistance $R_{ms} = \frac{v_o}{i_i} _{R_L=\infty}$ (V/A)	$R_i = 0$ $R_o = 0$

- Depending on the nature of the source signals and output loads, different types of amplifiers are needed:

- **Voltage** amplifier
- **Current** amplifier
- **Transconductance** amplifier
 - Voltage to current
- **Transimpedance** amplifier
 - Current to voltage

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Amplifier Frequency Response

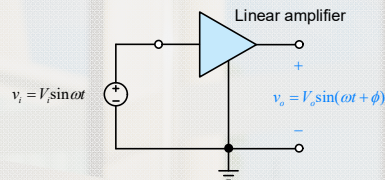


Fig. Measuring the frequency response of a linear amplifier

- When a sinusoidal signal is applied to a linear amplifier, the output is sinusoidal:
 - With **identical frequency** as input
 - But **different amplitude** and **phase**
- **Transfer function:** $T(\omega) = V_o(\omega) / V_i(\omega)$
- Amplitude response: $|T(\omega)| = V_o / V_i$
- Phase response: $\angle T(\omega) = \phi$

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Frequency Response

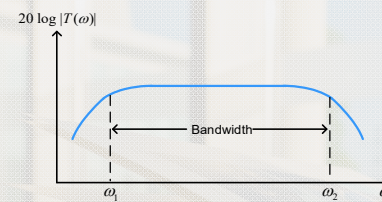


Fig. Typical magnitude response of an amplifier.

- **Bode Plot:** log-log plot of the **transfer function** vs. **angular frequency, ω**

- Vertical axis: $20 \log |T(\omega)|$
- Horizontal axis: $10 \log \omega$

- **Bandwidth**

- Band of frequencies over which the gain response falls by **3dB**

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Frequency Response of Low-Pass Filters

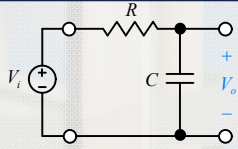


Fig. a low-pass network.

Transfer function:

$$T(\omega) = \frac{1/(j\omega C)}{R + 1/(j\omega C)} = \frac{1}{1 + j\omega RC} = \frac{1}{1 + j\omega/\omega_0}$$

$$\text{where } \omega_0 = \frac{1}{RC}$$

$$|T(\omega)| = \frac{1}{\sqrt{1 + (\omega/\omega_0)^2}}$$

$$\angle T(\omega) = -\tan^{-1}(\omega/\omega_0)$$

$$\omega_{3dB} = \omega_0$$

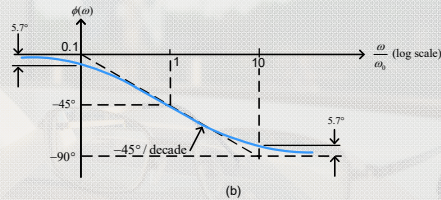
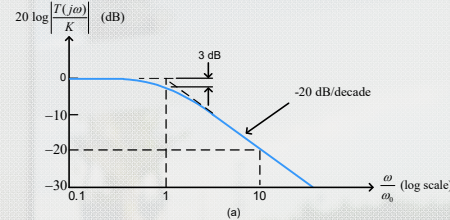


Fig. (a) Magnitude and (b) phase response of STC networks of the low-pass type.

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Frequency Response of High-Pass Filters

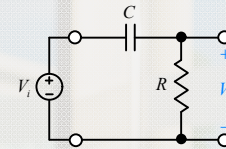


Fig. A high-pass network.

Transfer function:

$$T(\omega) = \frac{R}{R + 1/(j\omega C)} = \frac{1}{1 + 1/(j\omega RC)}$$

$$= \frac{j\omega RC}{1 + j\omega RC}$$

$$\text{Where, } \omega_0 = \frac{1}{RC}$$

$$|T(\omega)| = \frac{\omega/\omega_0}{\sqrt{1 + (\omega/\omega_0)^2}}$$

$$\angle T(\omega) = \tan^{-1}(\omega/\omega_0)$$

$$\omega_{3dB} = \omega_0$$

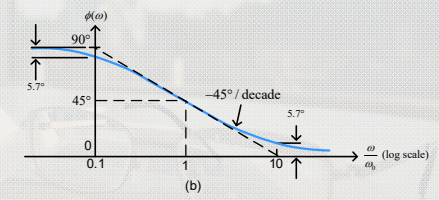
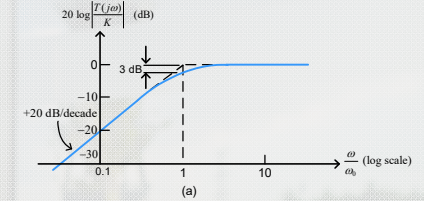
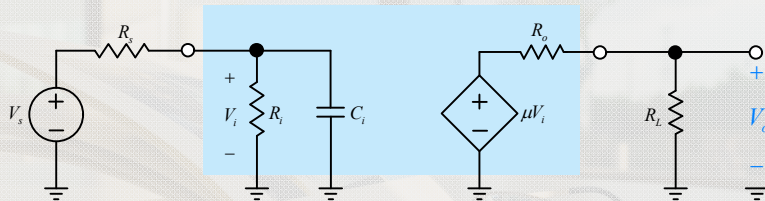


Fig. (a) Magnitude and (b) phase response of STC networks of the high-pass type.

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Example: Amplifier Frequency Response

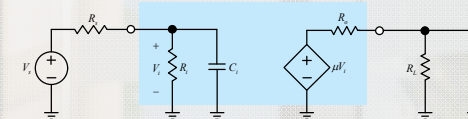
Given a voltage amplifier (input resistance R_i , input capacitance C_i , gain μ , output resistance R_o) driven by a source V_s with resistance R_s . A load R_L is connected to the output.



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Solution(a): Amplifier Frequency Response

- (a) Derive V_o/V_s as a function of frequency. Find dc gain and 3-dB frequency.



Solution:

$$(a) \quad V_i = V_s \frac{Z_i}{Z_i + R_s}$$

$$V_i = V_s \frac{1}{1 + R_s/Y_i} = V_s \frac{1}{1 + R_s[(1/R_i) + sC_i]}$$

$$\Rightarrow \frac{V_i}{V_s} = \frac{1}{1 + (R_s/R_i) + sC_i R_s}$$

$$\frac{V_i}{V_s} = \frac{1}{1 + (R_s/R_i)} \cdot \frac{1}{1 + sC_i \left(\frac{1}{R_s} + \frac{1}{R_i} \right)}$$

$$= \frac{1}{1 + (R_s/R_i)} \cdot \frac{1}{1 + sC_i [(R_s R_i)/(R_s + R_i)]}$$

$$V_o = \mu V_i \frac{R_L}{R_L + R_o}$$

$$\frac{V_o}{V_s} = \mu \frac{1}{1 + (R_s/R_i)} \cdot \frac{1}{1 + (R_o/R_L)} \cdot \frac{1}{1 + sC_i [(R_s R_i)/(R_s + R_i)]}$$

$$\tau = C_i \frac{R_s R_i}{R_s + R_i} = C_i (R_s || R_i)$$

$$T(j\omega) \equiv \frac{V_o}{V_s}(s) = \frac{K}{1 + \left(\frac{s}{\omega_0} \right)}$$

$$K \equiv \frac{V_o}{V_s}(s=0) = \mu \frac{1}{1 + \left(\frac{R_s}{R_i} \right)} \cdot \frac{1}{1 + \left(\frac{R_o}{R_L} \right)}$$

$$\omega_0 = \frac{1}{\tau} = \frac{1}{C_i (R_s || R_i)}$$

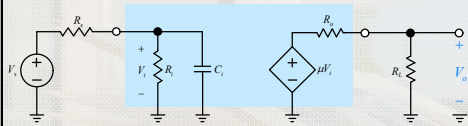
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Solution(b): Amplifier Frequency Response



- (b) Given $R_s = 20 \text{ k}\Omega$, $R_i = 100 \text{ k}\Omega$, $C_i = 60 \text{ pF}$, $\mu = 144 \text{ V/V}$, $R_o = 200 \Omega$, and $R_L = 1 \text{ k}\Omega$. Calculate the dc gain, and the 3-dB frequency.

Solution:



$$K \equiv \frac{V_o}{V_s}(s=0) = \mu \frac{1}{1 + \left(\frac{R_s}{R_i}\right)} \frac{1}{1 + \left(\frac{R_o}{R_L}\right)}$$

$$\omega_0 = \frac{1}{\tau} = \frac{1}{C_i(R_s \parallel R_i)}$$

(b)

$$K = 144 \frac{1}{1 + (20/100)} \cdot \frac{1}{1 + (200/1000)} = 100 \text{ V/V}$$

$$\omega_0 = \frac{1}{60 \text{ pF} \times (20 \text{ k}\Omega \parallel 100 \text{ k}\Omega)}$$

$$= \frac{1}{60 \times 10^{-12} \times [20 \times 100 / (20 + 100)] \times 10^3} = 10^6 \text{ rad/s}$$

$$f_0 = \frac{10^6}{2\pi} = 159.2 \text{ kHz}$$

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Solution(c): Amplifier Frequency Response



(c) Find $v_i(t)$ for each of the following inputs:

Solution: (c)

$$T(j\omega) \equiv \frac{V_o}{V_s}(j\omega) = \frac{100}{1 + j\left(\frac{\omega}{10^6}\right)}$$

(i) For $\omega = 10^2 \text{ rad/s}$,

$$|T| \approx 100 \text{ and } \phi = -\tan^{-1}10^{-4} \approx 0^\circ.$$

$$v_o(t) = 10 \sin 10^2 t \text{ V}$$

(ii) For $\omega = 10^5 \text{ rad/s}$,

$$|T| = 99.5 \text{ and } \phi = -\tan^{-1}0.1 = -5.7^\circ.$$

$$v_o(t) = 9.95 \sin(10^5 t - 5.7^\circ) \text{ V}$$

(iii) For $\omega = 10^6 \text{ rad/s}$,

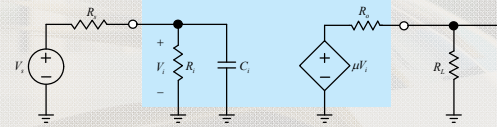
$$|T| = 100/\sqrt{2} = 70.7 \text{ V/V or } 37 \text{ dB and } \phi = -45^\circ.$$

$$v_o(t) = 7.07 \sin(10^6 t - 45^\circ) \text{ V}$$

(iv) For $\omega = 10^8 \text{ rad/s}$,

$$|T| \approx 1 \text{ and } \phi = -\tan^{-1}100 = -89.4^\circ.$$

$$v_o(t) = 0.1 \sin(10^8 t - 89.4^\circ) \text{ V}$$



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Typical Frequency Responses

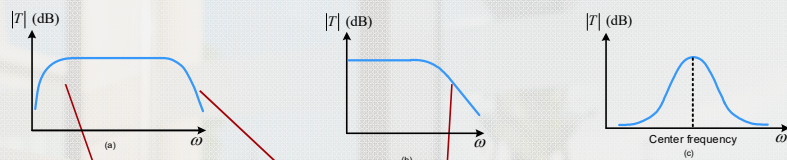


Fig. Frequency response for (a) a capacitively coupled amplifier, (b) a direct-coupled amplifier,

(c) a tuned or bandpass amplifier.

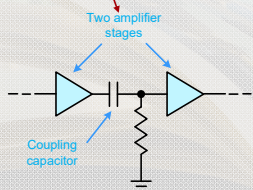


Fig. Use of a capacitor to couple amplifier stages.

High frequency cut-off due to **intrinsic capacitors** of the transistors

Low frequency roll-off due to **coupling capacitor**

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